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Applied Economics and Markets

Predicting Firm-Level Performance: Evidence from Italian Private Firms

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Academic Year 2025–2026

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Abstract

This thesis places itself in the firm dynamics literature by studying whether private firms' productivity changes are predictable in the Italian context. This matters because it can help clarify which variable categories carry predictive signal for firm productivity in this context, suggesting where future causal research might focus to explain large productivity shocks. Prior research has shown that productivity dynamics are non-Gaussian and heavy-tailed, with extreme expansions or contractions far more frequent than standard distributions imply. So, this thesis firstly assesses how productivity growth is distributed in the Italian context; secondly, it checks if productivity growth is predictable from balance-sheet and institutional data; thirdly, it determines if firms can be classified into the extreme tails of productivity growth distribution using available data; finally, it determines which categories of variables are most correlated with firm-level productivity outcomes in this environment. The results show us that in Italy the distribution of firm growth is non-normal and that productivity growth remains largely unpredictable; the picture is more nuanced regarding tails classification, firms about to experience exceptional positive productivity shocks are discriminable from balance-sheet characteristics, while bottom-tail collapses prove largely unpredictable. Finally, among the variables that do carry predictive power, Revenue per Capita, Asset Turnover, and the South Dummy emerge as the most consistent signals across all productivity measures; instead for tail classification, top-1% expansions carry the most consistent signal from ROA, Capital Turnover, and Firm Age, while bottom-1% collapses carry the most consistent signal from Log Sales and Labour Cost per Employee.¹

¹In the preparation of this thesis, use was made of generative artificial intelligence tools, specifically Claude Sonnet 4.6 (Anthropic, `claude.ai`, June 2026). Details are provided in Appendix C.

1. Introduction

1.1. Motivation and Research Question

A fundamental question in the firm dynamics literature is whether the large productivity shocks observed at the firm level are systematically correlated with observable firm characteristics, or whether they are largely idiosyncratic and orthogonal to anything a researcher can measure from balance sheets and institutional data. Documenting whether such correlations exist, and if so which categories of variables carry the most informational content, contributes to our understanding of firm heterogeneity in two ways. First, if certain types of firms (defined by their financial structure, operational efficiency, geographic location, or institutional context) are disproportionately associated with positive or negative productivity shocks, this indicates that such shocks are not purely idiosyncratic but they might be correlated with observable firm characteristics in a structured way. Second, if specific groups of variables are consistently associated with productivity outcomes across different measurement approaches and samples, this identifies the dimensions of firm behaviour that future, more directly causal research should prioritise, and offers a practical starting point for identifying which categories of firms are more likely to be approaching a large positive or negative shock, even without establishing why. If, on the other hand, the correlation between observable characteristics and firm-level shocks is weak across all variable categories, this is itself an important result, suggesting that the variance of firm performance is driven predominantly by idiosyncratic, unobservable events that no amount of observable information can capture.

Understanding the true difficulty and consequence of the productivity prediction problem requires looking closely at how company growth actually plays out in the real world. Changes in firm-level productivity are not normally distributed, but instead feature pronounced heavy tails where extreme expansions or contractions occur far more frequently than a normal distribution would imply (Stanley et al., 1996; Melcangi and Sarpietro, 2024). This distributional property makes the question of whether observable characteristics are correlated with extreme tail outcomes separately interesting from the question of whether they are correlated with average growth, since a model that explains the bulk of the distribution well may still have no ability to distinguish the firms that end up in the extreme tails, and vice versa. Understanding whether the correlation between observable characteristics and firm-level outcomes is stronger or weaker at the extremes of the distribution (and whether it is asymmetric between the top and bottom tails) therefore motivates the separate tail analysis alongside the overall-growth exercise.

Precedent literature and the existing evidence it has produced on firm performance predictability are overwhelmingly concentrated on large, highly transparent, publicly

listed U.S. corporations subject to strict accounting standards (see Melcangi and Sarpietro (2024)), while it has not studied private firms operating in different markets in other settings. Whether these dynamics apply to different institutional settings remains a large gap in the literature and this thesis tries to address this by shifting the focus to the Italian economy. Italy serves as an ideal setting because, despite being one of Europe's largest economies, its productive system is dominated by small and medium private enterprises. These firms are fundamentally different from large American public companies because they operate under Italian Civil Code accounting that has different accounting rules compared to US GAAP, and they have to make less disclosures compared to listed businesses, this makes them less transparent than them, and all of this makes the Italian private company context a useful and largely untested setting in which to check whether correlations documented for listed firms generalise, or are instead an artefact of the data-rich, highly disclosed environment in which they were first found. To tackle this challenge, this study enriches the predictive framework with a novel set of variables tailored specifically to the Italian corporate landscape. By incorporating Operational Efficiency metrics (to capture underlying management quality), Working Capital dynamics (to signal liquidity constraints and operational agility), and unique institutional arrangements such as industrial districts and the North-South divide (to account for regional agglomeration effects), this analysis aims to uncover correlations that previous studies that utilized standard publicly listed company data simply could not observe. It must also be noted that since in most countries of the world the number of listed companies is substantially lower than that of private firms (World Bank, 2024a,b), an analysis of the Italian firm landscape might bring up elements and characteristics that might be extendable to other nations where public companies are not as widespread as their private counterparts.

Given all of this, this thesis aims to answer four questions: how is private firms productivity growth distributed in Italy? Are idiosyncratic changes in productivity for Italian private firms actually predictable using observable balance-sheet characteristics and institutional variables? Do the extreme tails of the productivity growth distribution exhibit different and asymmetric predictability, and if so in which direction? And which specific categories of variables (such as Working Capital, Operational Efficiency, or Italian Institutional features) are most systematically associated with firm-level productivity outcomes in this environment? Answering these questions should help us better understand whether the fundamental unpredictability of firm dynamics extends to the Italian firm landscape or if this unpredictability was a limitation of previous frameworks that relied on public firm data.

1.2. Literature Review

The study of firm growth dynamics has a long history, originating with Gibrat (1931), whose Law of Proportionate Effect proposed that the rate of growth of a firm is independent of its size. This was the first systematic statement about the statistical properties of firm-level growth, and it implied that firm size follows a log-normal distribution over time. These assumptions dominated the empirical literature for decades, with many studies testing and partially confirming the law across different countries and time periods, as surveyed by Sutton (1997). The distributional properties of firm growth were examined more rigorously by Stanley et al. (1996), who showed that the distribution of firm growth rates is Laplace-like rather than Gaussian, with tails far heavier than Gibrat’s log-normal model would predict. This finding established that extreme positive and negative firm-level outcomes are far more frequent than classical models assume, and it opened the question of whether such outcomes are systematically related to observable firm characteristics or are instead fundamentally random. The macroeconomic relevance of firm-level growth dynamics was further documented by Davis et al. (2006), who highlighted a secular decline in both the cross-sectional dispersion of firm growth rates and average firm-level volatility, demonstrating that the distribution of firm-level outcomes is not static but evolves over time. Adding to this macroeconomic perspective, Decker et al. (2014) documented a long-term decline in U.S. business dynamism (specifically noting a drop in startup rates and employment reallocation) which underscores how closely aggregate economic health is tied to the shifting behavior of young and high-growth firms. At the micro level, Syverson (2011) documents that productivity differences across firms within the same industry are enormous and persistent, raising the natural question of whether these differences are themselves predictable from observable firm characteristics.

The question of whether firm-level performance changes are predictable from observable characteristics has received surprisingly little direct attention. Most of the firm dynamics literature has focused on documenting the distributional properties of growth and productivity rather than on testing their predictability (Sutton, 1997; Stanley et al., 1996). An important exception is Melcangi and Sarpietro (2024), who find that despite rich firm-level predictors, productivity growth for U.S. publicly listed firms remains largely unpredictable, and that the degree of nonlinear persistence means history matters asymmetrically. To rigorously evaluate whether this unpredictability is a universal feature of firm dynamics, this thesis adopts the core predictor categories established by Melcangi and Sarpietro (2024) (Financial Characteristics, Firm Characteristics, Competitors’ Performance, and Production Input Growth) and applies them to the private, SME-dominated economy of Italy. However, because the existing predictability literature relies on large, highly transparent public corporations subject to strict disclosure requirements, relying solely on this baseline framework is insufficient for

a different institutional setting. Therefore, this study expands the predictive model by introducing new categories of variables (Working Capital, Italian Institutional Setting, Operational Efficiency and Balance-Sheet Dynamics). By anchoring the analysis in the established baseline variables and enriching it with these novel dimensions, this thesis allows for a much more comprehensive evaluation of firm-level predictability.

1.3. Preview of Findings

The empirical analysis delivers four interconnected findings that together paint a coherent picture of firm dynamics in the Italian private sector. The first finding concerns the distributional properties of firm growth that tells us that productivity growth in the Italian private firm population is decisively non-Gaussian and heavy-tailed, a pattern that holds for sales growth as well, this confirms that extreme operational outcomes are a structural feature of Italian firm dynamics rather than statistical noise, and that any predictive framework must contend with a distribution where the most consequential events are also the rarest. The second finding is about baseline predictability, specifically, when evaluated on the standard set of balance-sheet predictors used in the existing literature, productivity growth proves largely unpredictable from observable firm characteristics, with out-of-sample R^2 values close to zero across all three productivity measures, namely Labour Productivity growth (a simple revenue-per-employee measure), Olley–Pakes Total Factor Productivity growth (a structural estimator correcting for simultaneity and selection bias), and Total Factor Productivity Shares growth (an index-number measure of total factor productivity); the same finding holds for sales growth, confirming it is not a measurement-specific result. Thus, this result holds even in the Italian private firm context, suggesting that the fundamental difficulty of predicting firm-level performance is not an artefact of data transparency or institutional setting but a more general property of firm dynamics. The third finding concerns predictability in the extreme tails of the growth distribution. The picture here is more nuanced than the overall-growth results suggest because top-1% positive shocks are strongly predictable from observable balance-sheet characteristics, reflecting that firms on the verge of exceptional expansions leave clear advance signals in their financial statements, while bottom-1% collapses prove largely unpredictable because they are driven predominantly by events not captured by accounting data. This asymmetry is informative about which categories of shocks leave an observable trace and which do not, a distinction that matters for any future work aimed at identifying firms approaching a negative shock, since it suggests the relevant signals may lie outside standard balance-sheet data. The fourth result concerns the contribution of novel predictors, enriching the predictor set with the new variable categories (Working Capital, Italian Institutional Setting, Operational Efficiency and Balance-Sheet Dynamics) yields a meaningful improvement in predictability and, even more importantly, the specific variables that

survive LASSO selection are economically interpretable: Revenue per Capita, Asset Turnover, and the South Dummy carry a consistent sign across all three productivity measures, while Labour Cost per Employee, Services Cost Growth, and the Depreciation Rate are each consistent across two of the three. The persistent negative coefficient on the South Dummy is particularly noteworthy, southern firms are associated with lower productivity growth and this holds across all three productivity measures rather than being specific to any single estimator. Tail classification reveals a different and largely asymmetric set of signals: bottom-1% collapses carry the most consistent signal from Log Sales and Labour Cost per Employee, both entering negatively, while top-1% expansions carry the most consistent signal from ROA, Capital Turnover, and Firm Age, all entering negatively, alongside a positive signal from firms actively scaling their cost base and accumulating capital through Services Cost Growth, Equity Growth, and Share Capital Growth.

1.4. Structure

The remainder of this thesis is organised as follows. Section 2 describes the data and presents descriptive statistics. Section 3 introduces the estimation methods (OLS, LASSO, Random Forest) and the cross-validation framework. Section 4 reports the prediction results. Section 5 concludes by examining the implications of the findings, the limitations of the exercise, and possible paths for future research.

2. Data and Sample Construction

2.1. Data Sources and Institutional Constraints

This analysis uses data from: AIDA database (Bureau van Dijk), EU KLEMS and ISTAT. The utilised AIDA dataset is composed of 14,330 active Italian firms over 2015–2024, cleaned of liquidating and bankrupt firms using legal status and liquidation procedure start date. To make sure no public companies entered the analysis, listed companies identified via Borsa Italiana registry were cross-checked against the sample and removed, confirming that all firms in the analysis are privately held. EU KLEMS National Accounts for Italy (2021) was used to construct the industry-year cost shares for the Total Factor Productivity estimation, with a NACE to ATECO mapping and with the 2021 data carried forward as a constant to fill the remaining 2022–2024 gap. ISTAT 2011 Industrial Districts list contained the 141 districts identified at the SLL (Local Labour System) level, used to construct the Industrial District dummy. The study uses the data to cover the four baseline categories of predictors commonly used in the literature, but it also contributes with over 20 context-specific predictors covering Operational Efficiency, Working Capital, Balance-Sheet Dynamics and Italian Institutional variables.

A central factor that shaped the way the analysis was carried out is that the Italian company structure directly affects the availability of data. Less than 500 companies are listed on the Italian stock exchange (Borsa Italiana, 2024). This makes the Italian situation fundamentally different from the American one where Compustat covers thousands of listed firms. This has an impact on the data because big firms are subject to stricter disclosure requirements (Repubblica Italiana, 1991, 2015), meaning that the bulk of Italian economic activity is less transparent in its reporting. This is very evident on AIDA database where bigger firms tend to disclose more information, while smaller ones file only the minimum required by law, shrinking the number of variables available for analysis. Thus this study had to focus more on the use of variables relating to balance sheet information and income statement items, which are the most consistently reported across firms of all sizes under Italian Civil Code obligations (Repubblica Italiana, 1942), rather than the more granular operational and market-based disclosures that would be available for listed firms.

Another fundamental element that shaped the analysis is the difference in principles between the American accounting system and the Italian one. More specifically in the U.S.A. expenses on income statements are indicated using a destination-of-expense method as mandated by U.S. GAAP (Financial Accounting Standards Board, 2023) (dividing costs into cost of goods sold, selling expenses and administrative expenses). Instead, the Italian accounting system under the Civil Code (Repubblica Italiana, 1942) uses the nature-of-expense method for the income statement (classifying costs by their nature for example: materials, personnel, services and depreciation). This structural difference makes many of the variables utilized in the standard predictability literature not available in AIDA, more specifically we cannot use:

- **Research and Development expenditure** is not indicated in the sheets as under Italian GAAP (Organismo Italiano di Contabilità, 2016) are either expensed within general costs or capitalised under intangible assets without their own dedicated disclosure line, making it impossible to isolate given also that since the companies are small they do not share additional documentation that might allow partitioning the general costs in their specific destinations;
- **Acquisitions/total assets** are not indicated;
- **KZ/WW/HP financial constraint indices** (Kaplan and Zingales, 1997; Whited and Wu, 2006; Hadlock and Pierce, 2010) are not available as that would require market capitalization data, which is unavailable to private firms;
- **Market-to-book ratio** is not available as it would require market prices, again unavailable for private firms;
- **Patent data** cannot be taken into account as often not available.

2.2. Predictor Set

All predictors and outcomes are drawn from the same firm-year observations. Stock and ratio variables are taken from the firm’s balance sheet for the year prior to the outcome, while growth predictors are computed as the log-change over the year before that. The predictor set is organised into two groups, one composed of four categories that follow the baseline variable structure previously used by Melcangi and Sarpietro (2024), and another four categories consisting of this paper’s context-specific variables.

The first of the categories is the one of **Financial Characteristics**, which captures the balance-sheet health and the financial position of the firm. This is relevant as financially constrained or distressed firms may respond differently to productivity shocks, a firm with high leverage or negative profitability has less capacity to invest in response to positive shocks and less buffer to absorb negative ones. The variables used for this category are: Return on Assets ($ROA = \text{Operating Income} / \text{Total Assets}$), a profitability dummy (that has value 1 if $ROA > 0$), Debt-to-equity ratio, and Equity-to-assets ratio.

The second category is **Firm Characteristics** whose objective is to help measure the structural features of the firm that affect baseline sales, productivity level and growth potential. The importance of this category is given by the fact that older firms tend to have more stable but less dynamic productivity trajectories, while bigger firms benefit from economies of scale but may be less agile, both age and size are well-documented predictors of growth rate heterogeneity in the firm dynamics literature, seen as early as in Dunne et al. (1989). This category’s variables are: Firm Age, Log Sales, and Log Total Assets.

The third category is named **Competitors’ Performance**, which measures the industry environment in which the firm operates. A firm’s productivity shock does not happen in a vacuum; wide demand shifts, input price movements or technological changes affect all firms in the sector simultaneously. By controlling for industry-level performance, we can isolate the truly idiosyncratic component of each firm’s shock. In this case the variables are: leave-one-out industry-year average of Sales Growth (the mean sales of all other firms in the same 2-digit ATECO sector and year, excluding the firm itself) and the leave-one-out industry-year average of Total Factor Productivity Shares (TFPS) of the other firms in the same sector.

The last of the baseline categories is **Production Input Growth**, whose objective is to capture changes in the firm’s factor inputs, which are directly linked to output growth through the production function. This is relevant to the analysis because input growth is mathematically correlated with output growth through the production function, so including it as a control allows the remaining variation in productivity growth to be interpreted as something beyond simple factor accumulation. The variables used to do this are the lagged growth rates of Wages, Materials, Personnel Costs, Tangible Fixed Assets, and Sales.

The first of the novel categories of variables introduced by this thesis is **Operational Efficiency and Capital Utilisation**. This category allows us to measure how productively a firm deploys its resources; a firm that generates more revenue per unit of invested capital or per employee is revealing information about its underlying productivity that is not captured by input levels or balance sheet stocks alone. Furthermore, the depreciation of tangible fixed assets extends this to capital vintage, as a firm with a high Depreciation Rate relative to its asset base is operating with older and more heavily used capital, which can be associated with lower productivity growth potential. Capital Turnover, Log Revenue per Capita, Asset Turnover and Log Value Added per Capita and Depreciation Rate are the variables composing this category.

Working Capital is the second novel category of our analysis, it is a relevant group of variables as Working Capital dynamics capture the liquidity cycle of the firm, that is how quickly it converts inventory into sales and collects receivables relative to how long it takes to pay suppliers. A firm with a long Commercial Cycle or high Inventory Days is tying up cash in operations, which constrains its ability to respond to productivity shocks and signals potential operational fragility. The variables of this category are: Commercial Cycle Duration in days, average Inventory Holding Period in days and Payables-to-Receivables ratio.

The third new group of variables of our analysis is the one of **Balance-Sheet Dynamics**. These matter as changes in the balance-sheet stocks over time reveal managerial decisions about recapitalization, investment and cost structure that are not visible from levels alone; a firm actively raising equity capital or growing its Operating Profit is on a different trajectory from one that is decapitalising or shrinking its service cost base. This group of variables regarding the Balance-Sheet Dynamics is formed by the growth rates of Equity, Net Profit, Operating Profit, Share Capital, and Services Costs, alongside the Log Labour Cost per Employee, Interest Coverage ratio and Intangible Ratio.

The last category of new variables introduced by this paper is the one treating the **Italian Institutional and Geographic Context**. These variables capture the dimensions of the Italian economic system that have no equivalent in Compustat. We have as variables: Export Status that captures exposure to international competition and the push to be more efficient that comes with that; the Group Size that captures access to internal capital markets within conglomerates; the South Dummy that captures the widely documented North-South productivity divide as shown by Iuzzolino et al. (2011); finally the Industrial District membership that captures agglomeration externalities specific to the Italian productive structure (Becattini, 1990).

The first four categories (on the left-hand side of Table 1) form the baseline predictor set previously used in the firm dynamics literature while the final four categories (on the right-hand side of Table 1) are novel contributions of this thesis.

Table 1: Complete predictor variable list

Variable	Definition/Measurement	Variable	Definition/Measurement
Financial Characteristics		Working Capital	
ROA	Operating income / Total assets	Commercial Cycle	DSO + Inv. days – DPO (days)
Profitable dummy	1 if ROA > 0, else 0	Inventory Days	Avg. inventory / (Revenue / 365)
Debt/equity ratio	Total debt / Equity	Payables/Receivables	Trade payables / Trade receivables
Equity-to-assets	Equity / Total assets		
Firm Characteristics		Italian Institutional	
Firm Age	Current – founding yr	Export Status	1 if Operatore Estero, else 0
Log Sales	$\ln(\text{Revenue}_{t-1})$	Group Size	$\ln(1 + \text{No. companies in group})$
Log Total Assets	$\ln(\text{Total assets}_{t-1})$	South Dummy	1 if province in Mezzogiorno
		Industrial District	1 if in ISTAT 2011 district
		District Specialisation	Set of 9 category dummies
Production Input Growth		Operational Efficiency	
Wages Growth	$\Delta \ln(\text{Wages}_{t-1})$	Capital Turnover	Revenue / Invested cap.
Materials Growth	$\Delta \ln(\text{Materials}_{t-1})$	Log Revenue per Capita	$\ln(\text{Revenue}/\text{Employees})$
Personnel Growth	$\Delta \ln(\text{Personnel}_{t-1})$	Asset Turnover	Revenue / Total assets
TFA Growth	$\Delta \ln(\text{TFA}_{t-1})$	Log VA per Capita	$\ln(\text{Value added}/\text{Employees})$
Sales Growth	$\Delta \ln(\text{Revenue}_{t-1})$	Depreciation Rate	Tangibles depreciation / TFA
Competitors' Performance		Balance-Sheet Dynamics	
Competitor Sales Growth	LOO industry-year mean $\Delta \ln(\text{Rev.})$	Equity Growth	$\Delta \ln(\text{Equity}_{t-1})$
Competitor TFP Growth	LOO industry-year mean TFPS growth	Net Profit Growth	$\Delta \ln(\text{Net profit}_{t-1})$
		Operating profit growth	$\Delta \ln(\text{Op. profit}_{t-1})$
		Share Capital Growth	$\Delta \ln(\text{Share capital}_{t-1})$
		Services Cost Growth	$\Delta \ln(\text{Services cost}_{t-1})$
		Log Labour Cost/Emp...	$\ln(\text{Labour cost}/\text{Employees})$
		Interest Coverage	Op. income / Financial charges
		Intangible Ratio	Intangible assets / Total assets

Note: Stock and ratio variables taken from the prior year; $\Delta \ln$ growth predictors computed as the log-change from the year before that. DSO = Days Sales Outstanding; DPO = Days Payable Outstanding; LOO = leave-one-out; TFA = Tangible Fixed Assets.

2.3. Outcome Set

Five prediction targets are studied: Labour Productivity growth, Olley–Pakes TFP growth, Total Factor Productivity Shares growth, sales growth, and sales levels. The first three of these targets are all productivity measures but they are not interchangeable. This is because Labour Productivity is calculated by dividing output by workers so it picks up input mix signal as much as true efficiency, Olley–Pakes TFP corrects for this but requires positive investment which substantially shrinks the sample, and Total Factor Productivity Shares sidesteps that restriction by using industry cost shares instead, meaning that any finding that holds across all three does not hinge on any single set of assumptions. Sales growth is included to separate internal efficiency from external market performance and to test whether low predictability is specific to productivity or a broader feature of firm dynamics, while sales levels serve as a falsification check since firm size is highly persistent and a framework that fails to predict levels trivially would call the entire pipeline into question. Each of the five targets will now be defined one after the other.

Labour Productivity growth (LP) is defined as the log-change in revenue per employee: $\Delta \ln(Y_t/L_t)$, where Y_t is revenue and L_t is the number of employees in year t . It requires only non-missing revenue and employees, giving it the largest estimation sample of 54,619 firm-year observations.

Olley–Pakes TFP growth (OP-TFP) is constructed using the structural estimator of Olley and Pakes (1996), which addresses two identification problems that cause OLS production function estimates to be biased. The first is a *simultaneity bias*: because firms observe their own productivity when choosing inputs, labour and capital are positively correlated with the unobserved productivity shock, causing OLS to attribute to inputs part of what is genuinely a productivity effect. The second is a *selection bias*: firms with low productivity are more likely to exit, so any sample that conditions on survival over-represents firms that drew high productivity realisations, inducing a further downward bias on the capital coefficient.

Olley and Pakes (1996) address both problems by exploiting the firm’s investment decision as a proxy for its unobserved productivity. If investment is strictly increasing in the firm’s productivity conditional on its capital stock and age, the investment equation can be inverted to express unobserved productivity as a function of observable investment, capital, and age, this inversion controls for simultaneity. Selection is controlled for separately by estimating the firm’s survival probability as a function of the same observables and including it as an additional control in the final stage. The labour coefficient is recovered in a first stage from a partially linear regression, and the capital coefficient is recovered in a third stage by exploiting the Markov structure of productivity and the timing assumption that capital is predetermined with respect to the current-period productivity shock. Specifically, productivity follows a first-order Markov process,

$$\omega_{it} = g(\omega_{i,t-1}) + \xi_{it}, \quad (1)$$

where ξ_{it} is an innovation that is mean-independent of capital k_{it} , since capital is chosen before ξ_{it} is realised. This orthogonality condition, combined with the non-parametric control for the lagged productivity state $\omega_{i,t-1}$ recovered from the first stage, identifies β_k without requiring external instruments. The resulting firm-level TFP level is:

$$\hat{\omega}_{it} = y_{it} - \hat{\beta}_l l_{it} - \hat{\beta}_k k_{it} - \hat{\beta}_a a_{it}, \quad (2)$$

where y_{it} , l_{it} , k_{it} , and a_{it} are log output, log labour, log capital, and Firm Age respectively.

The OP-TFP level is the residual from this procedure (output not explained by the estimated contributions of labour, capital, and age) and the growth series used as the prediction target is its first difference. Because the inversion of the investment equation is only valid when investment is strictly positive, the estimator requires positive revenue,

positive investment, positive employees, and non-missing Firm Age. These conditions reduce the sample to 10,152 firm-year observations, making this the most restrictive of the three productivity measures and selecting disproportionately towards larger, actively-investing firms.

TFPS (Total Factor Productivity Shares) is the preferred productivity measure of Decker et al. (2020), also used by Melcangi and Sarpietro (2024), who estimate the firm-level TFP as the residual of the revenue function:

$$z_{i,t} = y_{i,t} - \beta_c c_{i,t} - \beta_k k_{i,t}, \quad (3)$$

where $y_{i,t}$ is log revenue, $c_{i,t}$ is the log of a composite variable input, $k_{i,t}$ is the log capital stock, and β_c, β_k are industry-year average revenue elasticities constructed from cost shares in BLS KLEMS data. The key feature of this approach is that $c_{i,t}$ is a single bundled input -cost of goods sold (COGS) from Compustat- which by construction already combines materials, labour, intermediates, and energy into one line item. The elasticity β_c is therefore computed from the macro aggregate $(II + COMP)/GO$ from KLEMS, where II is intermediate inputs and $COMP$ is employee compensation, exactly matching the perimeter of the firm-level variable.

Applying this formula directly to Italian balance-sheet data is not straightforward, because Italian Civil Code accounts are organised *by nature* rather than *by destination*. While Compustat's COGS is a destination-based aggregate (grouping all costs attributable to goods sold regardless of their economic nature) Italian statutory accounts report costs in separate lines according to their type: raw materials (B.6), services (B.7), rentals (B.8), personnel costs (B.9), and inventory changes (B.11, A.2), among others. There is no single pre-constructed COGS equivalent. We therefore reconstruct the composite variable input algebraically so that its perimeter matches the macro elasticity:

$$M_{it} = \underbrace{(B.6_{it} + B.11_{it} - A.2_{it})}_{\text{materials consumed for goods sold}} + \underbrace{B.7_{it}}_{\text{services}} + \underbrace{B.8_{it}}_{\text{rentals}} + \underbrace{B.9_{it}}_{\text{personnel costs}}, \quad (4)$$

where B.11 adjusts raw-material purchases for the change in raw-material inventories and A.2 deducts the build-up of finished-goods inventories, so that only the input cost attributable to goods actually sold is captured. The full personnel cost block B.9 is included because the macro elasticity $\bar{\beta}_{st}^c$ is built from $(II + COMP)/GO$, and omitting labour from the firm-level variable would leave the entire labour contribution to value added inside the TFP residual and this causes a distortion especially for Italian firms where personnel costs typically represent the largest cost component.

The firm-level TFP measure is then computed as:

$$\ln A_{it} = \ln Y_{it} - \bar{\beta}_{st}^c \ln M_{it} - \bar{\beta}_{st}^k \ln K_{it}, \quad (5)$$

where Y_{it} is revenue, K_{it} is the tangible fixed asset stock, and $\bar{\beta}_{st}^c$ and $\bar{\beta}_{st}^k$ are industry-year average cost shares constructed from EU KLEMS National Accounts for Italy, with 2021 shares carried forward to cover 2022–2024. This approach avoids the need for firm-level input price data and does not require the strict filter conditions of the OP estimator. The TFPS estimation sample covers 31,522 firm-year observations.

Sales Growth is the log-change in total revenue: $\Delta \ln Y_t$. Unlike the productivity measures, it requires no structural estimation or identifying assumptions, making it a transparent benchmark: if even this directly observable measure of firm performance is unpredictable, the unpredictability of firm dynamics is confirmed regardless of measurement choice.

Sales levels ($\ln Y_t$) are predictable by construction due to firm-size persistence, and the near-unit root in firm size means that $\ln Y_{t-1}$ alone explains most of the variance in $\ln Y_t$. High R^2 for levels therefore provides no evidence of genuine predictive signal and serves as a ceiling benchmark.

For all growth targets, tail prediction is tested separately by binarising the growth distribution, this means that the bottom 1% indicator equals 1 for the worst-performing firms in the training distribution and 0 otherwise, and analogously for the top 1% and either tail. Thresholds are computed from the training data only and applied to the held-out fold, preventing leakage.

2.4. Descriptive Statistics

Before proceeding with descriptive statistics here is specified the structure of the dataset. The raw AIDA dataset covers 14,437 firms over the years 2015–2024. After dropping the non-active firms and after requiring non-missing revenue the number of active firms ranges from 6,333 in 2015 to a peak of 9,057 in 2023, with a slight decline to 8,842 in 2024. The observations for each growth series are: 66,917 for Sales Growth, 55,979 for Labour Productivity, 63,354 for capital growth, and 56,201 for employment growth; the panel is unbalanced, reflecting entry and exit over 2015–2024. Note that the Labour Productivity growth series of 55,979 observations is the raw count before merging with the predictor panel; after the merge the LP estimation sample used in the prediction exercise is 54,619 firm-year observations. Different measures of productivity were used in the analysis as a means of comparison for the TFPS that yields, that as specified earlier presents a total of 31,522 firm-year observations, more specifically, we have the previously mentioned Labour Productivity whose estimation sample yields 54,619 firm-year observations, we then have the Olley–Pakes TFP that instead has 10,152

observations. The differences in observations between these productivity measures are given by the fact that each one of them has different sets of requirements for the firms analysed, for instance the Olley and Pakes (1996) requires positive investment, positive revenue, and positive employees among others, these requirements exclude part of the firm observations from the analysed dataset.

The sample is dominated by small and medium enterprises: across the 6,098 firm-year observations in the estimation panel (2017–2024), median revenue is €1,613,182, median employees is 9, and median equity is €291,641. The size distribution is heavily right-skewed, specifically the mean revenue has a value of €11.7 million with a standard deviation of €96.4 million, and the 75th percentile of revenue (€4.74 million) remains well below the mean, confirming a long upper tail driven by a small number of large firms. Median tangible fixed assets stand at €199,504 and median wages at €211,022, consistent with a labour-intensive, asset-light SME population. The Debt/equity ratio has a median of 0.44 and a mean of 1.06, with the most extreme outlier reaching 285. It must be made present that, prior to imputation, this ratio has a substantially lower raw coverage of 4,346 observations (71.3%) compared to the other variables, because the ratio is undefined or misleading when equity is zero or negative, thus, in those cases, it was not directly observed. Mathematically, a firm’s equity is the difference between its Total Assets and its Total Liabilities, making the denominator of the ratio either undefined or misleading if the Total Liabilities exceed the Total Assets. As described below, the missing Debt/Equity observations (along with the small share of missing Wages and Group Size values) are median-imputed in Table 2 so that N is uniform at 6,098 across all variables, consistent with the imputation rule applied to the predictor matrix in the prediction exercise. Revenue is highly correlated with wages ($r = 0.87$) and with employees ($r = 0.83$), both reflecting firm-size co-movement rather than independent variation in factor intensities. Table 2 reports summary statistics computed on the same 6,098 firm-year observations (2,821 unique firms, 2017–2024) used as the estimation sample throughout the prediction exercise, so that N is identical across all nine variables.

The geographic distribution of firms mirrors Italy’s actual concentration of economic activity. The top provinces by firm count are Milano (693), Roma (676), Napoli (341), Torino (242), and Brescia (198). Of the 14,330 firms, 98% were successfully matched to a municipality via the ISTAT RACCORDO COMUNI mapping, and 23.3% of the sample (3,334 firms) fall within one of the 141 ISTAT 2011 industrial districts. The largest districts by firm count are Bergamo (270), Padova (211), Brescia (168), Como (141), and Prato (136). Export Status is rare, only 1.4% of firms are registered as active exporters (Operatore Estero), declining from 1.38% in 2016 to 0.89% by 2024, reflecting the prevalence of locally-oriented micro-firms. Start-up Innovativa and PMI Innovativa flags are near-zero at 0.2% each, confirming that their exclusion from the predictor set

Table 2: Summary statistics on the estimation sample ($N = 6,098$ firm-year observations)

	Mean	Std Dev	Min	p25	Median	p75	Max	N
Revenue (thousands of €)	11,664	96,422	1	677	1,613	4,740	5,852,126	6,098
Share capital (thousands of €)	455	6,737	0	10	20	100	500,000	6,098
Profit/Loss (thousands of €)	393	3,412	-47,931	5	32	154	133,513	6,098
Wages (thousands of €)	992	4,604	0	83	211	574	112,569	6,098
Employees	31.07	136.20	1	4	9	21	4,249	6,098
TFA (thousands of €)	2,097	15,667	1	48	200	862	614,543	6,098
Equity (thousands of €)	3,503	22,937	-17,175	83	292	1,199	815,558	6,098
Debt/Equity (%)	1.06	13.31	-631.85	0.14	0.44	0.97	285.18	6,098
Group Size	42.16	2,354.85	0	0	2	3	183,567	6,098

Note: TFA = Tangible Fixed Assets. Wages = Wages/Salaries. Group Size = number of companies in the corporate group. N = number of observations. Monetary values in thousands of euros. Wages (99.9% raw coverage), Debt/Equity (71.3% raw coverage, undefined when equity is zero or negative), and Group Size (99.8% raw coverage) are median-imputed for the small share of missing firm-years, using the same imputation rule applied to the predictor matrix in the LASSO/Random Forest estimation (Section 3), so that N is uniform across all variables.

is appropriate given the near-absence of variation.

Annual job creation rates range from 9.5% in 2020 (the Covid shock year) to 24.5% in 2019, while job destruction peaks at 23.9% in 2020. Entry and exit contribute meaningfully to gross job flows, specifically firm entry accounts for 2.1 to 4.8 percentage points of annual job creation, and firm exit for 2.0 to 6.3 percentage points of job destruction. Excess reallocation (that is the reallocation of jobs beyond what net employment growth requires) averages between 13% and 21% per year, indicating substantial turbulence in the employment distribution even in stable years. These figures underline that the Italian private firm population is far from static, and that the unpredictability of firm-level shocks documented in Section 4 occurs against a backdrop of significant underlying dynamism. The low number of employees, even in the higher percentiles, is the result of the widespreadness of micro-firms in Italy, as shown by OECD (2025), the number of business activities with only one employee are around 7% and 30% of the total employment, reaching heights of around 50% and 72% in manufacturing and non-financial services firms.

Table 3 reports the full annual job flow statistics.

Table 4 reports the coverage rates, that are defined as the fraction of non-missing observations, for all 43 predictor variables. The institutional and geographic variables (South Dummy, Industrial District, Export Status, Group Size) achieve the highest coverage at 80.8%, as they rely on firm-level registry information available for virtually the entire sample. Financial and operational variables show intermediate coverage between 33% and 54%, reflecting that smaller firms report fewer income statement and balance sheet details under Italian Civil Code minimums. The working capital variables (Commercial Cycle, Inventory Days, and Payables-to-Receivables ratio) have the lowest coverage at 21% to 34%, since they require both trade receivables and trade payables

Table 3: Annual job flow rates, Italian private firms (DHS specification with entry and exit)

Year	JC Rate	JD Rate	Net Growth	Reallocation	Excess Reallocation
2017	13.38%	7.69%	5.69%	21.07%	15.38%
2018	13.78%	6.75%	7.03%	20.53%	13.50%
2019	24.47%	8.20%	16.27%	32.68%	16.41%
2020	9.51%	23.91%	-14.40%	33.42%	19.02%
2021	15.85%	7.01%	8.84%	22.85%	14.01%
2022	11.89%	10.41%	1.48%	22.31%	20.83%
2023	12.09%	8.04%	4.05%	20.13%	16.08%
2024	13.28%	10.67%	2.61%	23.95%	21.34%

Note: JC = Job Creation rate; JD = Job Destruction rate. Reallocation = JC + JD. Excess Reallocation = Reallocation - |Net Growth|. Computed using the Davis et al. (1996) methodology on a balanced firm $\times$ year grid of 100,125 cells covering 11,125 unique firms. Entry and exit contributions included; internal gap transitions excluded.

to be reported simultaneously, a condition that many small firms do not meet. Missing values are handled via column-wise imputation with the training-set median prior to LASSO estimation, ensuring that coverage differences do not mechanically exclude firms from the analysis. The fact that key novel predictors survive LASSO selection despite this imputation pressure strengthens the idea that their signal is genuine rather than an artefact of sample composition.

Table 4: Predictor variable coverage rates (% non-missing observations)

Variable	Coverage	Variable	Coverage
Financial Characteristics		Working Capital	
ROA	54.0%	Commercial Cycle	20.9%
Profitable dummy	54.0%	Inventory Days	33.5%
Debt/equity	33.7%	Payables/Receivables	33.2%
Equity-to-assets	53.9%		
Firm Characteristics		Italian Institutional	
Log Sales	52.4%	Export Status	80.8%
Log Total Assets	54.0%	Group Size	80.8%
Age	80.7%	South Dummy	80.8%
		Industrial District	80.8%
		District spec. 1–9	80.8%
Production Input Growth		Operational Efficiency	
Wages Growth	36.8%	Capital Turnover	53.0%
Materials Growth	40.7%	Log Revenue per Capita	44.1%
Personnel Growth	38.0%	Asset Turnover	54.0%
TFA Growth	41.5%	Log VA per Capita	43.4%
Sales Growth	43.7%	Depreciation Rate	49.2%
Competitors' Performance		Balance-Sheet Dynamics	
Competitor Sales Growth	88.7%	Equity Growth	42.0%
Competitor TFP Growth	47.9%	Net Profit Growth	31.8%
		Operating profit growth	31.8%
		Share Capital Growth	45.1%
		Services Cost Growth	44.8%
		Log Labour Cost/Emp...	44.2%
		Interest Coverage	48.9%
		Intangible Ratio	53.9%

Note: Coverage computed on the full predictor panel of 128,970 rows prior to target merging. Missing values are imputed with the training-set median prior to LASSO and OLS estimation. District Specialisation dummies (9 categories) share a common coverage rate as they are derived from the same Industrial District mapping.

Three further methodological issues warrant explicit discussion: how entry and exit from the panel are handled, whether the resulting missingness and sample selection can be treated as random, and how zero or negative values are dealt with under the log transformations used throughout the predictor set.

Entry and exit. The AIDA panel is unbalanced by construction, with the number of active firms ranging from 6,333 in 2015 to a peak of 9,057 in 2023, this raises a concern for the predictability exercise because if exit is non-random and correlated with the predictors or the outcome, the estimation sample for any given year systematically excludes

the firms that performed worst (or, more rarely, were acquired after performing best), which would bias the cross-sectional relationship between observable characteristics and growth outcomes. Two features of the construction mitigate this concern without fully eliminating it. First, firms are only dropped from the analysis once their legal status indicates an active liquidation or bankruptcy procedure, so a firm experiencing a severe but non-terminal productivity or sales shock remains in the panel and continues to contribute observations in subsequent years; the panel does not artificially smooth over distress short of formal exit. Second, the job-flow statistics show that entry and exit account for a modest, stable share of gross job flows (entry contributing 2.1 to 4.8 percentage points of annual job creation, exit 2.0 to 6.3 percentage points of job destruction), indicating that the unbalanced panel reflects ordinary firm-level churn rather than an episodic wave of exits concentrated in any single year that could distort the cross-validation folds. The exception is 2020, where job destruction spikes to 23.9%, and a Covid-robustness check confirms that excluding this period does not materially change the productivity-predictability results.

Missingness and selection. Predictor coverage varies substantially across variables, from 88.7% for the leave-one-out Competitor Sales Growth measure down to 20.9% for the Commercial Cycle Variable, reflecting the tiered disclosure requirements of Italian Civil Code accounting, under which larger firms report considerably more line items than the minimum required of small firms. This pattern of missingness is unlikely to be missing completely at random, since it is correlated with firm size, but it plausibly satisfies the weaker missing-at-random condition once size and sector are accounted for: a firm’s decision to disclose, for example, trade receivables in sufficient detail to compute the Commercial Cycle is governed by statutory size thresholds rather than by the firm’s own knowledge of its forthcoming productivity shock. Missing values are therefore imputed using the training-fold median, computed separately within each cross-validation fold to avoid leakage, rather than dropped, since list-wise deletion under this coverage pattern would disproportionately remove smaller, less transparent firms and bias the estimation sample toward the largest end of the size distribution. A missingness robustness exercise tests this assumption directly by comparing the baseline imputed sample against a near-complete-case restriction requiring 80% non-missing predictor values; the resulting decline in out-of-sample R^2 (from 0.042 to 0.034) is consistent with, rather than contradictory to, the missing-at-random assumption, since it reflects the complete-case subsample skewing toward larger, more stable firms whose idiosyncratic shocks are inherently harder to predict, not a failure of the imputation procedure itself.

Zero and negative values under log transformation. Several predictors and outcomes in this thesis are constructed as natural logarithms or log-changes, which is undefined for zero or negative values, and Italian SME balance sheets frequently

contain both as firms can report zero revenue in a given year, zero or negative profit, and zero or negative equity in roughly a third of firm-years. Three approaches are used depending on the variable. For strictly positive flow variables such as revenue, wages, and tangible fixed assets, observations with non-positive values are treated as missing for the purposes of the corresponding log or log-growth predictor, rather than being assigned a small positive constant, since that would distort the magnitude of genuine small-firm observations relative to true zeros and was judged less transparent than allowing the standard median-imputation procedure described above to handle the resulting missingness. For variables that are naturally signed and where negative values are economically meaningful rather than reporting artefacts, such as Net Profit Growth and Operating Profit Growth, the underlying levels are differenced and the percentage growth is computed directly rather than via a log-difference whenever the sign changes between the two consecutive years used to compute the growth rate, since a log-difference is undefined whenever either period is non-positive and a firm moving from a loss to a profit (or vice versa) is precisely the kind of informative event that an undefined log-growth would discard. Finally, ratio-type predictors with an economically meaningful zero or negative range, such as the Debt/Equity ratio, are left in levels rather than logged.

3. Methodology

This section describes the statistical and machine learning methods used in the prediction exercise. Before presenting the estimators, it is worth clarifying what *predictability* means in this context, since the term is sometimes conflated with forecasting. Predictability, as used throughout this thesis, refers to the question of whether observable firm characteristics explain a statistically and economically meaningful share of the variance in productivity growth across firms, as assessed by out-of-sample R^2 under cross-validation. Forecasting, by contrast, concerns the construction of real-time predictions for genuinely unknown future outcomes. This thesis evaluates how much of the variation in firm-level productivity growth can be predicted by observable firm characteristics without making claims about the practical deployability of these models as live forecasting tools. Out-of-sample R^2 is the natural criterion for this exercise because it directly measures the proportion of outcome variance explained on data the model has never seen, penalising overfitting automatically.

The analysis compares three estimators, Ordinary Least Squares, LASSO, and Random Forest, chosen to span the spectrum from a simple linear baseline to a flexible non-linear model. Their roles are distinct and complementary: OLS provides an unregularised linear baseline; LASSO is the primary estimator, performing variable selection and revealing which predictors carry genuine signal; and Random Forest serves

as a test of whether meaningful non-linearities are present in the predictor-outcome relationship. All three are evaluated on identical held-out folds using 5-fold cross-validation, ensuring that comparisons across estimators and across the two predictor sets (the 14-variable benchmark set replicating Melcangi and Sarpietro 2024 and the full 43-variable extended set, both described in Section 2.2) are made on an equitable basis.

3.1. Measuring Non-Gaussianity

Before turning to the prediction methods, this section introduces the tools used to characterise the distributional properties of firm-level growth series, since the non-Gaussian, heavy-tailed nature of these series (documented in Section 2.4) is itself one of the central findings of this thesis and shapes how predictability is interpreted throughout. Four statistics and two formal tests are used.

Moment skewness measures the asymmetry of a distribution around its mean:

$$\text{Skew} = E \left[\left(\frac{X - \mu}{\sigma} \right)^3 \right], \quad (6)$$

where μ is the mean and σ is the standard deviation. A positive value indicates a longer right tail, consistent with a small number of firms experiencing large positive growth shocks. Because this measure relies on the third moment, it is highly sensitive to extreme outliers, which is precisely the feature of interest in this setting but also motivates the use of a more robust, quantile-based alternative.

Kelley skewness is a robust, quantile-based measure of asymmetry that does not depend on extreme moments:

$$\text{Kelley} = \frac{P_{90} - 2P_{50} + P_{10}}{P_{90} - P_{10}}, \quad (7)$$

where P_n denotes the n -th percentile of the distribution. Because it is built entirely from quantiles, Kelley skewness is far less sensitive to a handful of extreme observations than moment skewness, making it a useful cross-check in a setting where the tails themselves are the object of study.

Excess moment kurtosis measures tail heaviness relative to the Gaussian benchmark:

$$\text{Ex. Kurt} = E \left[\left(\frac{X - \mu}{\sigma} \right)^4 \right] - 3, \quad (8)$$

where the subtraction of 3 normalises the statistic so that a Gaussian distribution has an excess kurtosis of 0. Like moment skewness, this measure is highly sensitive to extreme values, which can make it unstable in finite, heavy-tailed samples.

Crow–Siddiqui kurtosis is the robust, quantile-based counterpart used as the

primary tail-heaviness measure throughout this thesis:

$$\text{C-S Kurt} = \frac{Q_{0.975} - Q_{0.025}}{Q_{0.75} - Q_{0.25}}, \quad (9)$$

which equals 2.91 under Gaussianity. Because it is constructed from quantiles rather than raw moments, it remains stable even when the underlying distribution has extremely heavy tails, and it is therefore the measure relied upon when comparing tail heaviness across growth series in Section 2.4.

In addition to these descriptive statistics, two formal tests of normality are used. The **Jarque–Bera test** jointly tests for the presence of skewness and excess kurtosis, with test statistic

$$JB = \frac{n}{6} \left(S^2 + \frac{K^2}{4} \right), \quad (10)$$

where S is the sample skewness, K is the sample excess kurtosis, and n is the sample size; under the null of normality, JB is asymptotically distributed as $\chi^2(2)$. The **Lilliefors test** is a Kolmogorov–Smirnov-type test for normality that does not require the population mean and variance to be known in advance, instead estimating them from the sample; this makes it more appropriate than a standard Kolmogorov–Smirnov test in the present setting, where the true mean and variance of each growth series are themselves unknown. Both tests are applied to each growth series in Section 2.4, and both reject the null of normality at the 1% level for every series considered.

Applied to this thesis’s data, all four growth series (Sales, Employment, Labour Productivity, Capital) are continuous variables by construction, since each is a log-difference of an underlying continuous balance-sheet quantity. This is precisely the setting in which moment-based skewness and excess kurtosis are most fragile since both statistics are built from third and fourth powers of deviations from the mean, so in a heavy-tailed sample a handful of extreme firm-year observations can dominate the entire statistic, making it noisy and sample-dependent rather than a stable description of the bulk of the distribution. This is the substantive reason the Crow–Siddiqui kurtosis and Kelley skewness, both built from quantiles rather than moments, are treated as the primary tail-heaviness measures throughout this thesis rather than as a secondary cross-check. The Jarque–Bera test inherits the same fragility, since it is constructed directly from the moment-based statistics; it is therefore used here only as a confirmatory, large-sample test of the null of normality (which it rejects overwhelmingly) rather than as a precise characterisation of the degree of non-normality, a role left to the quantile-based measures.

3.2. Prediction Methods

3.2.1 Ordinary Least Squares

Ordinary Least Squares (OLS) finds the coefficient vector that minimises the sum of squared residuals:

$$\hat{\beta}_{\text{OLS}} = \arg \min_{\beta} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}. \quad (11)$$

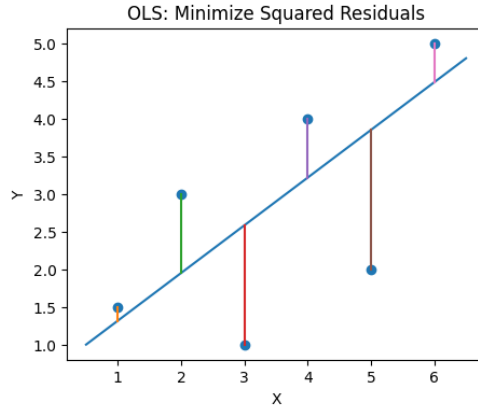


Figure 1: Geometric interpretation of OLS. Each observation (x_i, y_i) lies at a vertical distance $e_i = y_i - \hat{y}_i$ from the fitted line. OLS chooses the coefficient vector $\hat{\beta}_{\text{OLS}}$ that minimises the sum of squared lengths of these residual segments.

In this analysis OLS serves as an unregularised linear baseline against which LASSO and Random Forest are compared. It has no mechanism for variable selection and cannot capture non-linearities, so it is not expected to be the best-performing estimator; its role is to establish a lower bound on what a simple linear model can recover from the data. One formal issue is worth noting: the heavy-tailed growth distributions documented in Section 4.1, with Crow–Siddiqui kurtosis ranging from 5.9 to 11.5, mean that the normality assumption underlying exact t - and F -based inference does not hold. Since the evaluation criterion throughout is out-of-sample R^2 rather than coefficient significance, this does not affect the validity of the results; OLS predictions are evaluated on held-out data like any other estimator, with no distributional assumptions required.

OLS is applied to the continuous growth and level outcomes only. Tail prediction uses the Logistic-LASSO classifier described in Section 3.3.

3.2.2 LASSO

The Least Absolute Shrinkage and Selection Operator (Tibshirani, 1996) adds an ℓ_1 penalty to the OLS objective:

$$\hat{\beta}_{\text{LASSO}} = \arg \min_{\beta} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda \|\beta\|_1, \quad (12)$$

where $\lambda \geq 0$ controls the degree of regularisation. The geometry of the ℓ_1 constraint region drives many coefficients to exactly zero, performing variable selection automatically. The regularisation parameter λ is selected by 5-fold cross-validation within the training data via `LassoCV`, using only training observations.

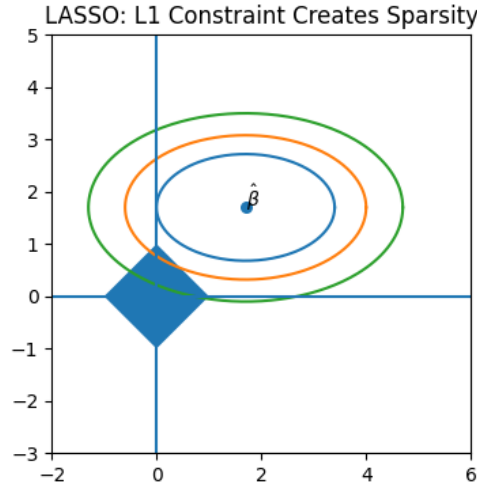


Figure 2: Geometric interpretation of LASSO, following Tibshirani (1996). The elliptical contours represent equal values of the residual sum of squares, centred at the unconstrained OLS estimate $\hat{\beta}$. The ℓ_1 constraint region $\{\beta : \|\beta\|_1 \leq t\}$ is a diamond with corners on the coordinate axes. The contours first contact this region at a corner, setting $\beta_1 = 0$ exactly and producing a sparse solution.

LASSO is the primary estimator for three reasons specific to this dataset. First, the 43-variable extended predictor set spans very different dimensions of firm performance, from financial ratios to binary institutional dummies, and automatic variable selection directly answers the thesis’s question of which characteristics carry signal. Second, many predictors have low coverage and are imputed from the training median; LASSO naturally shrinks attenuated signals toward zero, limiting the influence of imputed values on the results. Third, the predictor mix of continuous ratios, growth rates, and binary dummies creates a scale-sensitivity problem: without standardisation, the ℓ_1 penalty disproportionately shrinks variables with smaller natural variance regardless of their actual predictive content. Standardisation is therefore applied within each training fold before LASSO is fitted, as described in Section 3.4.

One limitation is directly relevant to interpreting the surviving predictors in Table 10: when predictors are correlated, LASSO tends to select one representative from each correlated group and shrink the rest to zero. Variables such as Revenue per Capita, Log Sales, and Log Total Assets are highly collinear in this dataset, so surviving predictors should be read as signals from a correlated cluster rather than as uniquely identified

drivers.

3.2.3 Random Forest

Random Forest (Breiman, 2001) aggregates the predictions of B decorrelated decision trees, each grown on a bootstrap resample of the training data:

$$\hat{f}_{\text{RF}}(\mathbf{x}) = \frac{1}{B} \sum_{b=1}^B T_b(\mathbf{x}), \quad (13)$$

where at each split only m randomly selected predictors are considered as candidates, which decorrelates the trees and reduces ensemble variance. In this analysis $B = 200$ trees are grown with $m = \lfloor \sqrt{p} \rfloor$ and a minimum-leaf size of 20 observations. The smaller-than-default m is chosen because several predictors are highly collinear, and restricting the candidate set at each split prevents any single dominant variable from being chosen at most nodes. The minimum-leaf constraint acts as a mild regularisation that limits overfitting on a sample with substantial heavy-tailed noise.

Random Forest can capture non-linear relationships and interaction effects that OLS and LASSO cannot, and it requires no standardisation since tree splits are invariant to monotonic transformations of the predictors. It is also not pulled by extreme values in the same way a squared-error linear model is, which is an advantage given the heavy-tailed distributions documented in Section 4.1.

Its main limitation in this context is that it cannot extrapolate beyond the range of outcomes observed in training: predictions for any observation are averages of leaf values from the training data. Since the firms of greatest substantive interest are precisely those in the extreme tails, Random Forest is structurally disadvantaged at exactly the task that matters most, and is therefore applied to the overall-growth targets only, with tail prediction handled by the Logistic-LASSO classifier described in Section 3.3.

3.3. Tail Classification

For the three productivity growth targets and Sales Growth, tail prediction is tested separately by binarising the growth distribution, the bottom 1% indicator equals 1 for the worst-performing firms in the training-fold distribution and 0 otherwise, and analogously for the top 1% and either tail. Thresholds are computed from the training-fold outcome distribution only and applied to the held-out fold, preserving the leakage-free protocol described in Section 3.4. Sales level is excluded from tail prediction, since the tails of a level target identify the smallest and largest firms rather than firms experiencing genuine productivity shocks.

The appropriate metric for this setting is the **Area Under the Receiver Operating Characteristic Curve** (AUROC). AUROC is defined as the probability that a randomly drawn positive observation (a tail-event firm) receives a higher predicted

score than a randomly drawn negative observation:

$$\text{AUROC} = \Pr(\hat{s}_i > \hat{s}_j \mid y_i = 1, y_j = 0), \quad (14)$$

where $\hat{s}_i$ is the model's predicted score for observation i . AUROC = 0.50 corresponds to random ranking and AUROC = 1.00 to perfect separation; values in the range 0.65–0.70 are regarded as economically meaningful for a rare-event screener, and values above 0.80 indicate strong discrimination. Crucially, AUROC is threshold-free: it evaluates the quality of the predicted *ranking* across all possible classification thresholds simultaneously, so it is unaffected by the base-rate imbalance that compresses OOS R^2 for rare events.

The classifier used for tail prediction is an **L1-penalised logistic regression** (`LogisticRegressionCV` from `scikit-learn`) with `class_weight='balanced'` and `solver='liblinear'`. The balanced class weight rescales the loss function so that each class contributes equally regardless of its frequency, preventing the classifier from degenerating to constant-zero predictions under severe imbalance. L1 penalisation is chosen to maintain consistency with the LASSO specification used for continuous targets: both implement ℓ_1 variable selection in the same predictor space, so differences in their outputs reflect the binary versus continuous outcome rather than differences in the regularisation family. The regularisation parameter C is selected by inner 3-fold cross-validation within each training fold. AUROC is averaged across the five outer held-out folds.

3.4. Cross-Validation and Model Evaluation

Model evaluation follows a **5-fold cross-validation** scheme. The firm-year panel is randomly partitioned into $K = 5$ non-overlapping folds of approximately equal size, repeated until every observation has appeared in the held-out fold exactly once. The reported metrics are averages across the five held-out folds:

$$\overline{R^2}_{\text{OOS}} = \frac{1}{5} \sum_{k=1}^5 R_k^2, \quad \overline{\text{RMSE}}_{\text{OOS}} = \frac{1}{5} \sum_{k=1}^5 \text{RMSE}_k, \quad (15)$$

where $R_k^2 = 1 - \frac{\sum_{i=1}^{n_k} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n_k} (y_i - \bar{y}_k)^2}$ is the coefficient of determination on the k -th held-out fold, with y_i the observed outcome, $\hat{y}_i$ the model prediction, $\bar{y}_k$ the mean of observed outcomes in that fold, and n_k the number of observations in that fold; and $\text{RMSE}_k = \sqrt{\frac{1}{n_k} \sum_{i=1}^{n_k} (y_i - \hat{y}_i)^2}$ is the root mean squared error on the same fold. Out-of-sample R^2 is the primary criterion as it is scale-free and directly interpretable as the share of outcome variance explained on held-out data. RMSE is reported alongside it as a complement expressed in the original units of the outcome variable.

The folds are formed by randomly shuffling firm-year observations across the entire

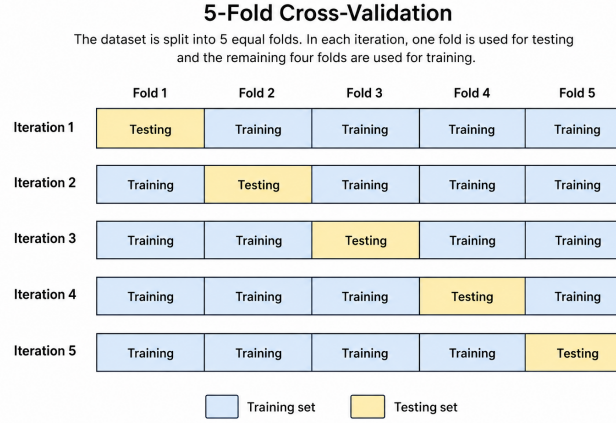


Figure 3: 5-fold cross-validation scheme. In each iteration, one fold serves as the held-out evaluation set (yellow) while the remaining four form the training set (blue). All preprocessing steps are fitted exclusively on the training folds to prevent data leakage.

panel. By treating each firm-year as an independent observation, this pooled cross-sectional approach ensures that every fold is fully representative of the underlying distribution of firm sizes, sectors, and macroeconomic conditions. This design directly aligns with the objective of the analysis that is to evaluate the general explanatory power of observable balance-sheet characteristics across the population of Italian private firms. The limitations of this approach are discussed in Section 5.

Leakage prevention is applied independently within each fold. After splitting, all preprocessing steps are fitted exclusively on the training fold and applied to the held-out fold:

1. *Missing value imputation:* a `SimpleImputer` with median strategy is fitted on the training fold. Missing values in both training and held-out folds are filled with the training-fold median, ensuring that no information from held-out observations enters the imputation.
2. *Winsorisation:* the 1st and 99th percentile bounds are computed from the training fold and clipped to both folds, preventing extreme values in the held-out fold from being driven by the held-out distribution.
3. *Standardisation:* a `StandardScaler` (zero mean, unit variance) is fitted on the training fold. The held-out fold is transformed using the training-fold mean and standard deviation.
4. *Tail thresholds:* the bottom 1% and top 1% cutoffs used to construct binary tail indicators are computed from the training-fold outcome distribution and applied to the held-out fold.

Applying these steps within each fold is critical. If, for example, the imputer were fitted on the full panel before splitting, the median used to fill missing values in the held-out fold would incorporate information from those very observations, creating a subtle form of leakage that would overstate out-of-sample performance.

No degrees-of-freedom correction is applied to the fold-average R^2 as the OOS generalisation gap already penalises model complexity implicitly: the model is evaluated on data it has never seen, so overfitting is punished directly through the prediction error on held-out observations.

Winsorisation is applied by clipping at the 1st and 99th percentiles of the training fold, with the bounds then applied to the held-out fold to prevent leakage. In this dataset, however, the observations being clipped are precisely the large expansions and collapses that the distributional analysis in Section 2.4 identifies as the most consequential events in the data. Winsorisation therefore necessarily tempers some of the signal the thesis is most interested in, in exchange for a more stable and comparable evaluation across folds. This means the reported out-of-sample R^2 values should be read as a measure of predictability for the winsorised distribution, which is very slightly more compressed than the raw series reported in Table 5.

LASSO hyperparameter selection uses an inner cross-validation loop that is entirely separate from the outer evaluation loop. Within each training fold, **LassoCV** applies 5-fold cross-validation to the training data alone to select the regularisation parameter λ that minimises out-of-fold MSE. The held-out evaluation fold has no role in λ selection.

3.5. Final Exercise Summary

The prediction exercise follows the framework and methods described in Section 3. Three productivity targets are studied (LP growth, OP-TFP growth, and TFPS growth) alongside two sales targets (Sales Growth and sales levels); all five targets are estimated on the same common intersection sample of 6,098 firm-year observations to ensure that comparisons across targets reflect differences in predictor signal rather than differences in sample composition. Tail prediction is tested separately for the bottom 1%, top 1%, and either tail of the growth distribution. Two predictor sets are compared throughout: the set of 14 variables replicating the Melcangi and Sarpietro (2024) structure (the M&S benchmark set) and the extended set of all 43 variables. The delta in out-of-sample R^2 between the two sets (for continuous growth targets) and the delta in AUROC (for tail classification) together quantify the contribution of the novel predictors introduced by this thesis.

3.6. Software

All analyses are implemented in Python 3, a general-purpose programming language widely adopted in both academic research and industry for its extensive ecosystem of scientific computing libraries. Three open-source libraries form the backbone of the analysis.

`pandas` (The Pandas Development Team, 2020) and `NumPy` (Harris et al., 2020) provide the foundational data structures and numerical routines used throughout: panel construction, variable transformations, winsorisation, and missing-value imputation are all performed with these tools.

`scikit-learn` (Pedregosa et al., 2011) is the primary machine learning library. It provides `LinearRegression` for OLS estimation, `LassoCV` for LASSO with automated inner cross-validation over the regularisation path, `RandomForestRegressor` for the ensemble estimator, and `KFold` for the outer cross-validation loop. `scikit-learn` is chosen to ensure that all three estimators are evaluated within an identical pipeline and that leakage-prevention steps (imputation, winsorisation, standardisation) are applied uniformly within each fold.

`statsmodels` (Seabold and Perktold, 2010) is used for the productivity estimation steps that require econometric routines beyond standard machine learning: specifically, the first-stage OLS regressions and probit models in the Olley–Pakes procedure, for which it provides the coefficient standard errors and specification diagnostics needed to validate the structural estimates.

4. Results

4.1. Productivity Growth distribution

A central finding of this thesis, consistent with Melcangi and Sarpietro (2024) and Stanley et al. (1996), is that all firm-level growth series are non-Gaussian and heavy-tailed, the relevant figures are reported in Table 5 showing the distributional statistics for the four main growth series. The Crow–Siddiqui kurtosis (a robust measure of tail heaviness that equals 2.91 under Gaussianity) substantially exceeds the benchmark for every series: Sales Growth reaches 7.96, employment growth 11.54, Labour Productivity growth 5.91, and capital growth 8.08. Both Jarque–Bera and Lilliefors tests reject normality for all series at the 1% level. The distribution of Sales Growth has a mean of 0.146 and standard deviation of 0.685, with moment skewness of 2.70 confirming a right-skewed distribution driven by a small number of very fast-growing firms. Employment growth shows the highest kurtosis at 11.54, reflecting the discrete nature of hiring and firing decisions that generate very frequent near-zero changes punctuated by large jumps.

Table 5: Distributional statistics of firm-level growth series

Series	Formula	N	Mean	Std	Skew	Kelley	C-S Kurt	Ex. Kurt
Sales Growth	$\Delta \ln Y_t$	66,917	0.146	0.685	2.700	0.243	7.954	42.732
Employment growth	$\Delta \ln L_t$	56,201	0.048	0.375	0.739	0.223	11.536	21.317
LP growth	$\Delta \ln(Y_t/L_t)$	55,979	0.066	0.564	1.722	0.070	5.914	39.646
Capital growth	$\Delta \ln K_t$	63,354	0.092	0.735	1.933	0.374	8.075	19.196
<i>Gaussian benchmark</i>					0	0	2.91	0

Note: C-S Kurt = Crow–Siddiqui kurtosis (equals 2.91 under Gaussianity); Skew = moment skewness; Kelley = Kelley skewness; Ex. Kurt = excess moment kurtosis. Jarque–Bera and Lilliefors tests both reject normality for all series at the 1% level. All series are log-changes between consecutive years.

Table 6 reports the conditional moments of residualised sales growth and TFPS changes, computed within quintiles of past sales. Heavy tails are present across all quintiles for both outcomes, the Crow–Siddiqui kurtosis of residualised Sales Growth ranges from 5.40 in the bottom quintile to 6.51 in the top quintile, and for TFPS from 3.57 in the top quintile to 5.80 in the bottom quintile. Non-Gaussianity is therefore not driven by any single region of the distribution but is a pervasive feature of firm dynamics across the entire performance spectrum. Kelley skewness in Sales Growth declines monotonically from +0.494 in the bottom quintile to -0.087 in the top quintile, indicating that firms with low past sales grow faster on average while already large firms face a more symmetric or slightly negatively skewed shock distribution, this is consistent with mean-reversion in firm size. TFPS exhibits the same pattern, with Kelley skewness falling from +0.470 in the bottom quintile to -0.306 in the top.

Table 6: Conditional moments by past sales quintile (residualised growth)

Past sales quintile	Sales Growth			TFPS growth		
	Std Dev	Kelley	C-S Kurt	Std Dev	Kelley	C-S Kurt
Q1 (bottom)	0.730	+0.494	5.401	0.257	+0.470	5.803
Q2	0.386	+0.169	5.607	0.182	+0.136	5.399
Q3	0.370	+0.055	5.833	0.152	-0.019	5.116
Q4	0.365	-0.003	5.917	0.224	-0.197	4.574
Q5 (top)	0.349	-0.087	6.506	0.340	-0.306	3.570

Gaussian benchmark: Kelley = 0, C-S Kurt = 2.91

Note: Growth series are residualised on ATECO 2-digit sector dummies and a linear time trend before quintile assignment. Quintiles are formed on lagged log-sales. C-S Kurt = Crow–Siddiqui kurtosis.

4.2. Prediction Results

On the base benchmark predictor set, productivity growth is largely unpredictable across all three measures, this is because as you can see in Table 7, LASSO out-of-sample R^2 on

the held-out test set is 0.042 for LP, 0.067 for OP-TFP, and 0.082 for TFPS, with OLS delivering almost identical values. We also have that LASSO selects between 11 and 14 of the 14 available variables across the three targets, meaning the low predictability is not a consequence of variable elimination but it is probably a reflection of the limited signal in the benchmark predictor set. Sales Growth on the benchmark set yields a LASSO out-of-sample R^2 of 0.117, equally modest. All of these results show us that despite a rich set of firm-level financial and structural predictors, the growth component of productivity is largely unpredictable, with sales growth showing a similarly modest signal.

Table 7: M&S benchmark predictor set: OLS and LASSO results

Target	N_{train}	N_{test}	OLS OOS R^2	LASSO OOS R^2	LASSO RMSE	Vars sel.
LP	4,879	1,219	0.041	0.042	0.450	13/14
OP	4,879	1,219	0.068	0.067	0.389	11/14
TFPS	4,879	1,219	0.081	0.082	0.228	14/14
Sales Growth	4,879	1,219	0.116	0.117	0.363	14/14
Sales level	4,879	1,219	0.943	0.943	0.361	12/14

Note: All metrics are averages over held-out folds from 5-fold cross-validation. OOS = out-of-sample. LP = Labour Productivity; OP = Olley–Pakes TFP; TFPS = Total Factor Productivity Shares.

Adding the novel predictors improves predictability across all targets, as seen in Table 8 and the delta comparison in Table 9. For LP, the LASSO out-of-sample R^2 rises from 0.042 to 0.097 (delta +0.056, RMSE falling from 0.450 to 0.438). For OP it rises from 0.067 to 0.102 (delta +0.035, RMSE from 0.389 to 0.381). For TFPS the improvement is more modest, from 0.082 to 0.098 (delta +0.017), suggesting that the TFPS structural estimator already absorbs much of the firm-level variation that the novel variables would otherwise capture. The mean delta out-of-sample R^2 across the three productivity measures is +0.036. For Sales Growth, the extended set raises LASSO out-of-sample R^2 from 0.117 to 0.144 (delta +0.027, RMSE from 0.363 to 0.358).

To test whether the low predictability of productivity growth is a linear-model artefact, we complement the LASSO results with a Random Forest (RF) estimator applied to the extended predictor set.² As reported in the *RF OOS R^2* column of Table 8, RF achieves 0.137 for LP, 0.163 for OP, and 0.152 for TFPS, improvements of roughly 4–6 percentage points over the corresponding LASSO estimates. These gains are moderate, while RF captures additional non-linear structure and interaction effects, the bulk of the signal remains inherently low, confirming that the ceiling on productivity growth predictability is a genuine economic feature of the data rather than

²Random Forest is trained using the same 5-fold cross-validation protocol as LASSO, with 200 trees, `max_features=sqrt`, and `min_samples_leaf=20`. Feature importances are averaged across the five folds.

a modelling limitation. The one exception is sales levels, where RF OOS R^2 of 0.894 falls below the LASSO figure of 0.945. This may happen because firm size persistence is a linear signal, and RF cannot extrapolate beyond the range of training values, making it structurally weaker than LASSO at exactly the target where predictability is driven by near-unit-root level dynamics rather than non-linear interactions.

Table 8: Extended predictor set: OLS, LASSO, and Random Forest results

Target	N_{train}	N_{test}	OLS OOS R^2	LASSO OOS R^2	LASSO RMSE	RF OOS R^2	Vars sel.
LP	4,879	1,219	0.094	0.097	0.438	0.137	24/43
OP	4,879	1,219	0.100	0.102	0.381	0.163	24/43
TFPS	4,879	1,219	0.099	0.098	0.226	0.152	33/43
Sales Growth	4,879	1,219	0.146	0.144	0.358	0.236	34/43
Sales level	4,879	1,219	0.944	0.945	0.355	0.894	27/43

Note: OLS and LASSO metrics are averages over held-out folds from 5-fold cross-validation. RF OOS R^2 is evaluated using the same 5-fold cross-validation as OLS and LASSO, with 200 trees, `max_features=sqrt`, and `min_samples_leaf=20`. OOS = out-of-sample. LP = Labour Productivity; OP = Olley-Pakes TFP; TFPS = Total Factor Productivity Shares.

Table 9: Delta OOS R^2 and RMSE: extended minus M&S benchmark (LASSO), and RF vs LASSO gain (extended)

Outcome	OOS R^2 M&S	OOS R^2 Ext.	Δ OOS R^2	RMSE M&S	RMSE Ext.	Δ RMSE	RF Δ vs LASSO
LP growth	0.042	0.097	+0.056	0.450	0.438	-0.013	+0.040
OP growth	0.067	0.102	+0.035	0.389	0.381	-0.007	+0.061
TFPS growth	0.082	0.098	+0.017	0.228	0.226	-0.002	+0.054
Sales Growth	0.117	0.144	+0.027	0.363	0.358	-0.005	+0.092
Sales level	0.943	0.945	+0.002	0.361	0.355	-0.006	-0.051
<i>Mean</i> (Δ prod. only)			+0.036			-0.007	+0.052

Note: Δ OOS R^2 and Δ RMSE are LASSO extended minus LASSO M&S benchmark (5-fold CV). RF Δ vs LASSO = RF OOS R^2 (extended, 5-fold CV) minus LASSO OOS R^2 (extended, 5-fold CV). Positive values indicate improvement. Mean over LP, OP, TFPS only.

The values of ΔR^2 across the three productivity measures (LP gains +0.056, OP +0.035, TFPS +0.017) is itself an informative pattern that requires special attention, and the reason is that since the TFPS is constructed by subtracting from log-revenue an industry-year-weighted combination of material input and labour cost via the EU KLEMS macro elasticities (equation 5), meaning that the residual already nets out the components of firm-level variation that flow through these channels, when predictors like Revenue per Capita and Labour Cost per Employee are then added to the model, they are partly duplicating information that has already been removed from the dependent variable, so they have less new variation left to explain and their estimated contribution shrinks. Labour Productivity does not perform this subtraction beforehand (it is just raw revenue per employee), so the same predictors carry their full unused signal when applied to it; OP, instead, sits in between, because it uses an investment-based proxy to control for productivity but does not directly subtract input costs themselves. The fact

that TFPS leaves so little room for the novel predictors to improve fit is therefore not a weakness of these predictors, it is direct evidence that the TFPS estimator captures more of the structural input-driven component of revenue variation than the other two measures, and that what remains in its residual is closer to the truly idiosyncratic productivity shock that the firm dynamics literature is ultimately trying to model.

4.3. Predictor Signals for Productivity Prediction

The novel predictors that survive LASSO selection consistently across productivity measures carry economically interpretable signals, specifically we have Table 10 that shows us that: Labour Cost per Employee and Revenue per Capita enter with the largest negative coefficients for LP and OP, consistent with firms with high revenue or labour cost per worker showing slower subsequent productivity growth; Log Value Added per Capita enters negatively for LP and OP but positively for TFPS, the only variable with this kind of reversal; the South Dummy enters negatively for all three measures, indicating that southern firms are associated with systematically lower subsequent productivity growth even after accounting for all other observable characteristics, a pattern consistent with the Italian North-South productivity divide; Capital Turnover and Asset Turnover both enter negatively across all three measures; Services Cost Growth and Interest Coverage enter positively for LP and OP but are shrunk to zero for TFPS; Industrial District and Intangible Ratio both enter positively across all three measures, while Depreciation Rate enters negatively for OP and TFPS but is shrunk to zero for LP, with a negative coefficient associated with firms operating older and more heavily depreciated capital where the variable survives.

Table 10: Novel predictors surviving LASSO (overall growth, $|\text{coef}| > 0$)

Predictor	LP coef.	OP coef.	TFPS coef.	Sign
Labour Cost per Employee	−0.0619	−0.0448	—	Negative (LP, OP)
Revenue per Capita	−0.0599	−0.0475	−0.0260	Negative (all)
Log VA per Capita	−0.0378	−0.0195	+0.0119	Mixed
South Dummy	−0.0224	−0.0166	−0.0049	Negative (all)
Capital Turnover	−0.0214	−0.0332	−0.0122	Negative (all)
Services Cost Growth	+0.0208	+0.0115	—	Positive (LP, OP)
Interest Coverage	+0.0100	+0.0011	—	Positive (LP, OP)
Industrial District	+0.0090	+0.0010	+0.0034	Positive (all)
Intangible Ratio	+0.0082	+0.0068	+0.0004	Positive (all)
Share Capital Growth	+0.0067	+0.0055	—	Positive (LP, OP)
Net Profit Growth	+0.0060	—	—	Positive (LP)
Asset Turnover	−0.0045	−0.0066	−0.0116	Negative (all)
Group Size	+0.0040	+0.0003	—	Positive (LP, OP)
Commercial Cycle	+0.0037	—	—	Positive (LP)
Inventory Days	−0.0013	—	−0.0069	Negative (LP, TFPS)
Depreciation Rate	—	−0.0082	−0.0258	Negative (OP, TFPS)

Note: Coefficients from LASSO on the extended predictor set, overall growth target, fitted on the last of the 5 cross-validation training folds. “—” indicates the variable was shrunk to exactly zero by LASSO for that target. District Specialisation dummies are omitted for brevity; they survive for LP (spec_6, spec_8), OP (spec_3, spec_6, spec_8), and TFPS (spec_8).

4.4. Tail Classification

Table 11 reports the Logistic-LASSO AUROC for all four growth targets, three tail types, and both predictor sets. The results differ sharply from what the overall-growth R^2 figures would suggest. The data contain a strong and robust ability to discriminate which firms will experience top-1% positive shocks as AUROC reaches 0.869 for LP, 0.886 for OP, 0.902 for TFPS, and 0.933 for Sales Growth on the extended predictor set (Table 11), values that point to strong and consistent discriminative power across all four targets. The ability to identify the *bottom* 1% is considerably weaker, with AUROC ranging from 0.655 to 0.721 across productivity measures and 0.639 for Sales Growth, though still meaningfully above random. The either-tail target sits between the two, between 0.751 and 0.769 for productivity and 0.751 for sales.

This top-versus-bottom asymmetry, already visible in the OOS R^2 figures but far more pronounced in AUROC, suggests that firms heading towards top-1% positive productivity or sales shocks tend to be associated with observable balance-sheet signals such as the scaling of personnel and material inputs, building sales momentum, and favourable leverage profiles, all captured by the Production Input Growth and Working Capital variables in the extended predictor set. The AUROC gains from adding the novel predictors are largest precisely for LP (delta +0.086 on the bottom tail, +0.058

on the top), where these operational signals have their full unrestricted effect, and smallest for TFPS (+0.027 bottom, -0.030 top), where the structural estimator has already netted out much of the input-driven component, consistent with the pattern in the overall-growth results above.

Firms heading towards bottom-1% collapses, by contrast, are overwhelmingly affected by events that leave no advance trace in accounting data like the loss of an anchor customer, a regulatory or litigation shock, the departure of key personnel, or a supply-chain rupture concentrated in a single product line. These events are largely unrecorded by the financial and operational variables observable through AIDA, which is why bottom-tail AUROC, while above random, remains substantially below top-tail AUROC across all four targets. This asymmetry is itself an interesting finding for firm dynamics research: the firm characteristics associated with exceptional growth are not simply the mirror image of those associated with collapse, and identifying which group of firms is approaching a positive shock appears to be a fundamentally easier task than identifying which is approaching a negative one. If this pattern were to be confirmed in other settings, it would have a relevant downstream implication for credit-risk practice, since lenders care disproportionately about the bottom tail, where defaults and losses concentrate (Das et al., 2007; Duffie et al., 2009), yet it is exactly that tail where balance-sheet information provides the least discriminative power, suggesting that closing this informational gap is likely to require non-balance-sheet signals such as payment behaviour, court filings, or customer-concentration data, rather than a richer set of accounting variables alone.

Table 11: Tail classification: Logistic-LASSO AUROC (5-fold CV)

Target	Tail	AUROC		
		S&M Benchmark	Extended	Δ (Ext. – Bench.)
LP	Bottom 1%	0.635	0.721	+0.086
	Top 1%	0.810	0.869	+0.058
	Either tail	0.705	0.758	+0.053
OP	Bottom 1%	0.660	0.704	+0.045
	Top 1%	0.865	0.886	+0.021
	Either tail	0.737	0.757	+0.019
TFPS	Bottom 1%	0.628	0.655	+0.027
	Top 1%	0.932	0.902	–0.030
	Either tail	0.766	0.769	+0.003
Sales Growth	Bottom 1%	0.650	0.639	–0.011
	Top 1%	0.908	0.933	+0.024
	Either tail	0.713	0.751	+0.038

Note: AUROC is averaged across held-out folds from 5-fold cross-validation. Classifier: L1-penalised logistic regression (`class_weight='balanced'`, `solver='liblinear'`, regularisation C selected by inner 3-fold CV). Bottom/Top 1% are binary targets equal to 1 if the firm falls in the respective tail of the training-fold shock distribution. Either tail = 1 if in either the bottom or top 1%. Sales level omitted as tail prediction is not meaningful for a level target. AUROC = 0.50 indicates random discrimination; AUROC = 1.00 indicates perfect separation.

4.5. Predictor Signals for Tail Classification

The Logistic-LASSO coefficients for the tail classification exercise are reported in Tables 12–14 for the bottom-1%, top-1%, and either-tail targets on the extended predictor set, and they reveal a pattern that is both consistent with and sharply asymmetric relative to the continuous-growth results of Section 4.3: Log Sales enters negatively for all three targets in the bottom-1% panel, consistent with larger firms being substantially less likely to experience catastrophic productivity collapses; Labour Cost per Employee enters negatively across all three targets in both the bottom-1% and either-tail panels, associating firms with a more structured and compensated workforce with a lower likelihood of collapse; Sales Growth enters positively for all three targets in the bottom-1% panel, suggesting that firms that recently grew fast are more likely to collapse next; Revenue per Capita enters positively for all three targets in the bottom-1% panel but negatively for LP and OP in the top-1% panel, possibly indicative of mean reversion from both ends of the productivity distribution simultaneously, a pattern invisible in the continuous LASSO where the two effects cancel; the South Dummy enters positively for all three targets in the bottom-1% panel, consistent with southern firms being not only slower-growing in expectation but disproportionately represented in the collapse tail; ROA enters negatively for all three targets in the top-1% panel,

consistent with a ceiling effect in which already-profitable firms are associated with a lower probability of top-1% expansions; Capital Turnover enters negatively for all three targets in the top-1% panel, with a particularly large coefficient for TFPS (-1.440), consistent with a ceiling effect for already-efficient firms; Firm Age enters negatively for all three targets in the top-1% panel, with older firms less likely to experience exceptional positive shocks; Services Cost Growth, Equity Growth, and Share Capital Growth all enter positively for all three targets in the top-1% panel, consistent with firms that are actively scaling their cost base and accumulating capital being more likely to achieve top-1% expansions; Intangible Ratio enters positively for LP and OP in the top-1% panel, consistent with intangible investment signalling future growth potential; and Equity Growth is the only variable that enters positively in both the bottom-1% and top-1% panels, suggesting that a sharp increase in equity capital flags heightened volatility in either direction rather than a specifically positive or negative trajectory.

Table 12: Logistic-LASSO coefficients for tail classification (extended predictor set, leading predictors by absolute magnitude): Panel A, Bottom 1%

Predictor	LP	OP	TFPS	Sign
Log Sales	-1.847	-0.130	-0.321	Negative (all)
Log VA per Capita	$+0.836$	$+0.580$	—	Positive (LP, OP)
Competitor TFP Growth	-0.623	-0.254	—	Negative (LP, OP)
Asset Turnover	$+0.609$	$+0.019$	$+0.026$	Positive (all)
Equity Growth	$+0.501$	$+0.046$	—	Positive (LP, OP)
Sales Growth	$+0.492$	$+0.342$	$+0.115$	Positive (all)
Export Status	-0.483	-0.060	—	Negative (LP, OP)
Revenue per Capita	$+0.450$	$+0.262$	$+0.292$	Positive (all)
Labour Cost per Employee	-0.427	-0.292	-0.285	Negative (all)
Profitability dummy	-0.352	-0.355	-0.188	Negative (all)
South Dummy	$+0.329$	$+0.164$	$+0.003$	Positive (all)

Table 13: Logistic-LASSO coefficients for tail classification: Panel B, Top 1%

Predictor	LP	OP	TFPS	Sign
Capital Turnover	-0.142	-0.323	-1.440	Negative (all)
ROA	-0.616	-0.684	-0.222	Negative (all)
Labour Cost per Employee	-0.571	-0.405	—	Negative (LP, OP)
Revenue per Capita	-0.491	-0.465	—	Negative (LP, OP)
Firm Age	-0.234	-0.332	-0.316	Negative (all)
Asset Turnover	—	-0.321	-0.172	Negative (OP, TFPS)
Services Cost Growth	$+0.207$	$+0.284$	$+0.122$	Positive (all)
Equity Growth	$+0.209$	$+0.264$	$+0.104$	Positive (all)
Intangible Ratio	$+0.203$	$+0.146$	—	Positive (LP, OP)
Competitor TFP Growth	$+0.165$	$+0.076$	$+0.068$	Positive (all)
Share Capital Growth	$+0.118$	$+0.083$	$+0.173$	Positive (all)

Table 14: Logistic-LASSO coefficients for tail classification: Panel C, Either tail

Predictor	LP	OP	TFPS	Sign
Labour Cost per Employee	−0.607	−0.498	−0.256	Negative (all)
Log Sales	−0.117	−0.173	−0.870	Negative (all)
Log VA per Capita	+0.336	+0.322	+0.061	Positive (all)
Profitability dummy	−0.199	−0.290	−0.272	Negative (all)
Firm Age	−0.186	−0.287	−0.067	Negative (all)
Equity Growth	+0.131	+0.158	+0.025	Positive (all)
Sales Growth	+0.116	+0.208	+0.032	Positive (all)
Services Cost Growth	+0.044	+0.070	+0.151	Positive (all)
South Dummy	+0.052	+0.038	+0.057	Positive (all)
Share Capital Growth	+0.045	—	+0.115	Positive (LP, TFPS)

Note: Coefficients from Logistic-LASSO fitted on the full dataset with inner 3-fold CV selecting C (`class_weight='balanced'`, `solver='liblinear'`). A positive coefficient increases the predicted probability of belonging to that tail. “—” indicates the variable was shrunk to exactly zero. Leading predictors by absolute magnitude shown; district specialisation dummies omitted for brevity. Predictor set: extended (43 variables).

4.6. Discussion of Findings

Having reported the empirical results for each of the four research questions in turn, this section steps back to ask what the findings mean, individually and together.

On the distribution of firm growth. That productivity and sales growth in the Italian private firm population are decisively non-Gaussian (Section 4.1) and this itself is informative about the nature of this economy. A heavy-tailed growth process means the typical firm-year is uneventful, but a small number of firm-years account for a disproportionate share of aggregate reallocation, consistent with the job-flow statistics reported earlier in Section 2. For policy, this implies that aggregate outcomes in the Italian SME sector are driven less by the median firm’s gradual evolution and more by a small set of extreme expansions and contractions, so interventions aimed at the average firm are likely to have a more diffuse effect than interventions aimed at identifying and supporting or restraining the firms in the tails. The fact that heavy tails are present across every quintile of past firm size (Table 6), rather than concentrated among small or young firms, further suggests that tail risk here is not a small-firm problem to be solved by growth into larger size classes, but a structural feature of doing business in the Italian private sector regardless of scale.

On overall predictability. The near-zero predictability documented in Section 4.2 speaks directly to the firm-dynamics motivation set out in the introduction: it tells us that, in this population, observable balance-sheet characteristics correlate only weakly with subsequent productivity growth and sales growth alike, however comprehensively mined. This does not mean balance-sheet information is uninformative, since the novel predictors introduced by this thesis raise predictability modestly, and the variables that survive selection are interpretable, but it does mean that the variance of firm-

level productivity growth and sales growth in this population contains a substantial component that is, at least with the data available here, irreducible, in the sense that no amount of additional observable financial-statement detail appears able to capture. That this conclusion replicates the finding of Melcangi and Sarpietro (2024) for large, transparent, publicly listed U.S. firms is itself an important result as it suggests that firm-level unpredictability is not an artefact of poor disclosure, small firm size, or a specific national accounting regime, but it is probably a more general property of firm dynamics that persists across very different institutional settings.

On predictability in the extreme tails. The motivation for studying tail predictability separately from overall growth was that the correlation between observable characteristics and extreme outcomes is of independent interest for firm-heterogeneity research, since identifying which firms are heading toward a rare, extreme shock is a conceptually different exercise from predicting the size of productivity growth across the full sample, and a model can succeed at one without succeeding at the other. The classification results in Section 4.4 reveal a sharp asymmetry, present for productivity and sales growth alike: top-1% positive shocks are strongly predictable from observable balance-sheet characteristics, with AUROC reaching 0.869–0.933 across the four targets on the extended predictor set, while bottom-1% collapses are only weakly predictable, with AUROC between 0.639 and 0.721. This asymmetry is informative in itself: firms approaching exceptional positive growth events are associated with a clear footprint in the balance sheet, namely accumulation of fixed assets, scaling of personnel and material inputs, and favourable leverage, while firms approaching collapse are not distinguished by any comparably strong balance-sheet pattern, consistent with such events being more often associated with factors the balance sheet does not record, such as the loss of an anchor customer, a litigation outcome, the departure of key personnel, or a supply-chain rupture concentrated in a single product line. This suggests that improving bottom-tail prediction in this population would likely require a different category of information altogether, plausibly payment behaviour, supplier and customer concentration, or legal and litigation records, rather than a richer set of financial-statement ratios, however many are added to the existing ones.

On which variables carry predictive power. The variables that survive LASSO selection most consistently (Section 4.3) cluster around Operational Efficiency and the Italian North-South institutional divide, rather than around the financial-characteristics or production-input-growth categories that form the baseline predictor set from the existing literature. This suggests that the novel contribution of this thesis, extending the predictor set beyond what is available for listed U.S. firms, is not merely additive but different in kind, as the predictors that travel from the Compustat-based literature carry comparatively little independent signal once richer, Italy-specific variables are available. The persistence of the South Dummy’s negative coefficient, in particular, is

consistent with the North-South divide being not only a levels phenomenon but also a growth-rate phenomenon: southern firms are associated with lower productivity growth conditional on their starting point, and this pattern holds across all three productivity measures and both predictor sets. Given the collinearity among predictors discussed throughout, however, these patterns should be read as describing which correlated groups of variables carry signal, not as identifying uniquely causal drivers.

Taken together. Each of the four findings is, on its own, a variation on a single theme as documented the heavy-tailed distribution sets a ceiling that the predictability, tail-predictability, and variable-selection results each confirm from a different angle. What the four findings establish jointly that none establishes alone is that this ceiling is a property of the data-generating process rather than of any particular modelling choice made in this thesis. A single low R^2 could plausibly be blamed on a poorly chosen predictor set or an unlucky sample; four independent findings, using different targets, different predictor sets, and a more granular tail-classification exercise, converging on the same conclusion is much harder to attribute to any one specification decision. The robustness exercises reported in Appendix A reinforce this further, showing that the ceiling does not move materially under alternative samples or imputation strategies. The joint reading therefore strengthens the conclusion of Melcangi and Sarpietro (2024): firm-level performance in the Italian private sector, much like in the publicly listed U.S. economy, appears to be governed by a substantially irreducible component of idiosyncratic risk, structurally similar across two economies that differ enormously in firm size, ownership structure, disclosure regime, and accounting convention.

5. Conclusion

This thesis set out to determine whether the productivity of Italian private firms is predictable from observable balance-sheet and institutional characteristics, using sales growth as a benchmark outcome to assess the generality of the findings. The answer is, for the most part, no: firm-level growth in this population is heavy-tailed and largely unpredictable, a fact that holds across three productivity measures and two predictor sets. Two genuine exceptions to this picture stand out and are worth restating in closing. First, the asymmetry between the top and bottom tails of the growth distribution is real, economically meaningful, and runs in a specific direction: positive shocks are strongly discriminable from balance-sheet data while negative shocks are not, an asymmetry that would matter for any future application aimed at flagging firms approaching distress, since it is precisely the harder-to-predict tail that such applications care most about. Second, a small set of novel, Italy-specific predictors, in particular Revenue per Capita, Asset Turnover, and the South Dummy, carry a stable and interpretable signal across all three productivity measures, suggesting that observable characteristics are not entirely

uninformative, just modestly so.

Contribution to the literature. Beyond the specific findings discussed in Section 4.6, this thesis makes three contributions of a more general nature. First, it is, to the knowledge of the author, among the first studies to test the predictability of firm-level performance focusing on private, non-listed firms rather than on large, transparent, publicly listed corporations, which matters because private firms make up the overwhelming majority of economic activity in Italy and in most other economies worldwide. Second, it extends the standard predictor set with over twenty context-specific variables covering Operational Efficiency, Working Capital dynamics, balance-sheet growth, and Italian Institutional features such as the North-South divide and Industrial District membership, none of which have an equivalent in the destination-of-expense accounting systems. Third, it combines three estimators of increasing flexibility, OLS, LASSO, and Random Forest, within a single cross-validation framework applied consistently across every target and predictor set, which allows the predictability ceiling, or the lack of one, documented throughout this thesis to be attributed to the data rather than to the choice of any single model.

Limitations. Several limitations of this analysis should be acknowledged. The AIDA panel is unbalanced, and predictor coverage is lowest for the working-capital variables, ranging from 21% to 34%, meaning that imputation may attenuate some coefficients and true predictability in a fully observed sample could be marginally higher than reported here. The sample period, 2015 to 2024, includes the 2020 Covid shock, which the job-flow statistics confirm was a major disruption to the Italian private firm population, and results may differ over a longer or more stable window, although the robustness check reported in Appendix A suggests this is unlikely to be a major concern for the headline conclusions. Finally, family ownership, a distinctive institutional feature of the Italian productive system, could not be operationalised as a predictor given the snapshot nature and widespread unavailability of ownership data in AIDA, leaving one of the most characteristic features of Italian private firms unexamined in this analysis.

Extensions. These limitations point toward natural directions for future work. Richer data sources, such as administrative tax records, payment data, or credit registry information, could substantially increase predictor coverage and quality, particularly for the working-capital variables that suffer most from missingness in AIDA. Extending the analysis to other European countries with similar civil-law accounting traditions would permit a direct test of which Italian Institutional variables carry signal that is Italy-specific and which instead reflect broader continental patterns. Incorporating text-based predictors from annual reports, news coverage, or court filings alongside the balance-sheet variables used here could push the predictability frontier further while testing directly whether the informational limits documented in this thesis are a property of the data sources used rather than of firm dynamics themselves. More

broadly, the central lesson of this thesis is that the next meaningful gain in firm-level predictability is unlikely to come from richer balance-sheet ratios alone, but from new categories of information that capture the kinds of idiosyncratic, firm-specific shocks that a financial statement is not really designed to record.

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Appendix

A. Predictability Robustness Checks

Two robustness exercises test whether the near-zero predictability finding of Section 4 is fragile to specification choices made along the way: the inclusion of the Covid period in the TFPS sample, and the imputation strategy used to handle low-coverage predictors.

Table A1 reports the full distributional comparison between the TFPS estimates computed on the full 2016–2024 sample and the truncated pre-Covid 2016–2021 sample. The near-identical statistics confirm that the 2020 Covid shock and the subsequent recovery years do not materially distort the structural TFP estimates.

Table A1: TFPS robustness: full sample vs pre-Covid truncation

Sample	N	Mean	Median	Std Dev	Kelley Skew	C-S Kurt
Full sample (2016–2024)	38,284	0.714	0.583	0.600	0.363	2.884
Truncated (2016–2021)	24,543	0.702	0.579	0.602	0.352	2.892
Difference (full – trunc.)		+0.012	+0.004	−0.002	+0.011	−0.007

Note: Kelley Skew = Kelley skewness (0 under Gaussianity). C-S Kurt = Crow–Siddiqui kurtosis (2.91 under Gaussianity). All differences are below 2% in relative terms, confirming robustness of the TFP estimates to the inclusion of Covid-period observations.

Several predictor variables have coverage as low as 21% (Working Capital variables). The baseline analysis imputes missing values with the training-set median. Table A2 reports three robustness checks testing whether the baseline results are driven by this imputation strategy, all using LP overall growth and the S&M benchmark predictor set.

Table A2: Missingness robustness: LP overall growth, S&M benchmark predictor set

Check	N	Features	LASSO OOS R^2
Baseline (full N , median imputation)	6,098	14	0.042
A: $\geq 50\%$ row coverage	6,098	14	0.042
B: $\geq 80\%$ row coverage (near complete-case)	6,003	14	0.034
C: Only predictors with $\geq 40\%$ coverage	6,098	11	0.041

Note: Check A restricts to firm-years where at least 50% of predictor values are non-missing before imputation. Check B uses the near-complete-case subsample ($\geq 80\%$ non-missing). Check C drops the three Working Capital variables and debt-to-equity ratio (all $< 40\%$ coverage) and re-runs on the remaining 11 variables. The stability of OOS R^2 from baseline through Checks A and C (0.042, 0.042, and 0.041) confirms that median imputation does not distort predictability. Check B’s decline suggests; if anything, fully observed firms are *harder* to predict, likely because they are larger and more stable firms where idiosyncratic shocks are more genuinely random. The stability of Check C confirms that the three low-coverage Working Capital variables do not materially drive the baseline result.

Read together, the two checks confirm that the unpredictability ceiling documented

in the main text is not an artefact of any single specification choice. Restricting TFPS to the pre-Covid window leaves its distributional statistics essentially unchanged, and restricting the LP benchmark sample by predictor coverage leaves out-of-sample R^2 essentially unchanged: 0.042 under a 50% coverage threshold and 0.041 after dropping the four lowest-coverage variables outright, with the one exception (Check B, a near-complete-case subsample) declining to 0.034 for reasons of sample selection rather than imputation failure, since that subsample skews toward larger, more stable firms whose shocks are inherently harder to predict. Whether the analysis is run on the full panel or a Covid-free window, with median imputation or with low-coverage variables removed entirely, the order-of-magnitude conclusion is unchanged: the unpredictability ceiling is a property of the data, not of the pipeline.

B. Structural TFP Estimation Parameters

Table B3 reports the structural parameter estimates underlying the Olley–Pakes and TFPS productivity measures described in Section 3.

Table B3: Structural TFP estimation parameters

Estimator	Parameter	Estimate
Olley–Pakes (Stage 1)	Labour elasticity ($\hat{\beta}_l$)	0.6186
Olley–Pakes (Stage 3)	Capital elasticity ($\hat{\beta}_k$)	0.1858
Olley–Pakes (Stage 3)	Age coefficient ($\hat{\beta}_a$)	0.4177
TFPS	Avg. variable input elasticity ($\bar{\beta}_c$)	0.7853
TFPS	Avg. capital elasticity ($\bar{\beta}_k$)	0.2147

The Olley–Pakes labour and capital elasticities sum to 0.804, below one, consistent with mild decreasing returns to scale or with capital and labour not fully spanning the firm’s input bundle in this specification; the age coefficient of 0.418 is positive but modest, indicating that older firms are somewhat more productive conditional on capital and labour, consistent with the small age effects typically reported in the Olley–Pakes literature. The TFPS elasticities, by contrast, sum to exactly one by construction, since they are derived from industry-year cost shares under the assumption of constant returns to scale rather than being freely estimated; the higher weight on the variable input elasticity (0.785) relative to capital (0.215) reflects the labour-intensive nature of the median Italian private firm documented in Table 2.

C. Declaration on the Use of Generative Artificial Intelligence

In accordance with the University of Bologna Policy on the Responsible and Ethical Use of Generative Artificial Intelligence (Università di Bologna, 2024), the following declaration is provided regarding the use of generative AI tools in the preparation of

this thesis.

- **Tool and version:** Claude Sonnet 4.6, a large language model developed by Anthropic, accessed via `claude.ai`.
- **Period of use:** May 2026 – June 2026.
- **Modes of use:**
 - Revision and optimisation of text to improve clarity and expositional coherence.
 - Support in presenting the statistical methods used in the analysis (OLS, LASSO, Random Forest, AUROC).
 - Assistance in writing and debugging Python code for the statistical analysis, in particular the implementation of a cross-validation pipeline for the tail classification exercise.
 - Grammatical and stylistic proofreading of the English text.
- **Sections affected:** the contribution was most concentrated in Section 3 (Methodology, in particular the subsections on OLS, LASSO, Random Forest, and cross-validation), with lighter involvement across the remaining sections in the form of text revision and proofreading.